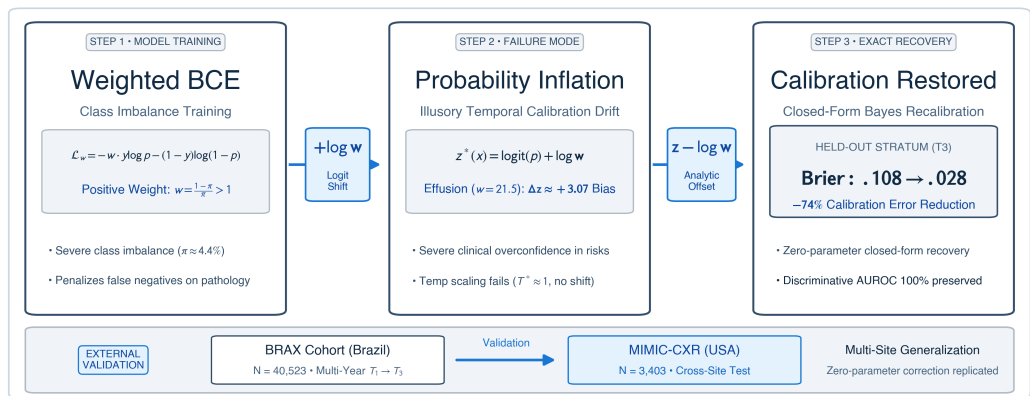


Graphical Abstract

Positive-Class Weighting Induces Systematic Probability Distortion in Deep Learning Models for Chest Radiography under Deidentified-Date Cohort Shift

Vishesh Panghal



Highlights

Positive-Class Weighting Induces Systematic Probability Distortion in Deep Learning Models for Chest Radiography under Deidentified-Date Cohort Shift

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- Positive loss weighting induces an additive population logit offset of $\log w$.
- Finite neural network logits empirically cluster tightly around theoretical $\log w$.
- Temperature scaling cannot correct this intercept-type calibration bias.
- Analytic offset ($z - \log w$) cuts 73.7% of Brier error without retraining.
- Selective triage exposes trade-offs between calibrated risk and screening referral.

Positive-Class Weighting Induces Systematic Probability Distortion in Deep Learning Models for Chest Radiography under Deidentified-Date Cohort Shift

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Abstract

Background and Objective: Clinical deep learning models for chest radiography frequently experience severe probability miscalibration under distribution shift. While positive-weighted binary cross-entropy (BCE) is standard practice to mitigate acute class imbalance, its mathematical distortion of output logit distributions has been conflated with feature drift. This study investigates the algebraic mechanism of loss-induced logit distortion, evaluates why standard post-hoc calibration fails, and formulates a zero-retraining recalibration and selective deferral pipeline.

Methods: We conducted a controlled 2×2 factorial experiment using two deep convolutional neural network architectures (DenseNet-121, ResNet-50) and two loss functions (unweighted vs. positive-weighted BCE) across three independent random seeds (12 models total) on 40,523 chest radiographs from BRAX. Models were trained on an anchor deployment stratum and evaluated across subsequent deidentified-date strata (T_1, T_2, T_3) with strict patient isolation. Recalibration strategies—temperature scaling, Platt scaling, and an $O(1)$ closed-form analytic logit offset ($z - \ln w$)—were evaluated alongside patient-clustered bootstrap ($B = 2000$). External validity was evaluated on MIMIC-CXR-JPG (3,403 radiographs, 289 patients).

Results: Positive-weighted BCE mathematically induces an additive population logit shift of $\ln w$, which temperature scaling structurally cannot resolve because scaling modifies logit dispersion without adjusting the intercept. In

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empirical finite networks, logit differences clustered tightly around theoretical values (mean $\Delta z = 3.29 \pm 0.81$ vs. theoretical 3.07 for Pleural Effusion). On stratum T_3 , the zero-parameter analytic offset reduced Brier score by 73.7% ($\Delta\text{Brier} = -0.0795$ [95% CI: $-0.0893, -0.0697$], $0.1079 \rightarrow 0.0284$) and reduced intercept error from $\alpha = -2.96$ to -0.45 without validation tuning. On MIMIC-CXR-JPG, discrimination was pathology-dependent (AUROC 0.688 for Effusion), with analytic offset reducing Brier error for Cardiomegaly and Pneumonia. Uncertainty-based selective deferral reduced automated error rates to < 0.01 at 70% coverage.

Conclusions: Severe probability miscalibration in chest radiograph classification across held-out deidentified-date strata is largely an artifact of objective function weighting rather than intrinsic dataset drift. The closed-form analytic offset provides an immediate, constant-time $O(1)$ algorithmic recalibration that restores probability reliability across held-out deidentified-date cohorts without model refitting.

Keywords: Chest radiography, Probability calibration, Class imbalance, Distribution shift, Selective prediction, Clinical decision support

1. Introduction

Deep learning models for chest radiography achieve high discrimination in retrospective benchmarks [1, 2]. However, prospective clinical deployment introduces substantial distribution shifts as patient demographics, imaging protocols, and disease prevalence evolve [3, 4, 5].

While discriminatory ranking (AUROC) exhibits gradual longitudinal attenuation, clinical decision support relies fundamentally on *calibrated posterior probabilities*: $P(Y = 1 \mid X = x)$ [6, 7]. Miscalibrated probabilities distort decision cutoffs, degrade expected utility, and compromise automated triage. Although recent studies report rapid “temporal calibration decay” in frozen neural networks [8, 9], thoracic datasets simultaneously exhibit severe class imbalance (prevalence $< 5\%$) routinely addressed via positive-weighted binary cross-entropy (BCE) [1, 10].

We demonstrate that these phenomena are deeply entangled: perceived temporal calibration collapse is largely an artifact of uncorrected training offsets under prevalence shift [11, 12]. Training with positive weight $w > 1$ mathematically shifts Bayes-optimal logits by an additive $+\ln w$ [13, 14]. Standard temperature scaling [15] rescales logit variance (z/T) but cannot

adjust this intercept. Consequently, weighted models over-predict positive probabilities, inducing severe Brier score and Expected Calibration Error (ECE) inflation that compounds as prevalence fluctuates.

1.1. Related Work

Probability Calibration and Class Imbalance: Deep neural networks frequently exhibit over-confidence [15]. While post-hoc recalibration techniques (temperature scaling, Platt scaling) are standard [16, 6], they were formulated for balanced settings. Piccininni et al. [17] showed that addressing class imbalance via resampling distorts calibration, which plug-in corrections can recover. In logistic regression, King and Zeng [13] proved that choice-based sampling adjustments induce an intercept shift of $\ln w$. Caplin et al. [14] extended this to loss-weighted deep learning, proving that class weighting distorts raw posteriors and recovering undistorted likelihoods on a static task. However, their analysis was confined to single static splits, leaving unexamined how loss-induced logit shifts behave under sequential cohort shifts, why temperature scaling fails, and whether post-hoc recalibration survives cross-institutional transfer.

Distribution Shift and Selective Prediction: Temporal dataset shift encompasses alterations in features $P(X)$, conditionals $P(Y | X)$, and label priors $P(Y)$ [3, 12]. Alexandari et al. [11] established that bias-corrected calibration adapts effectively under label shift. Selective classification routes ambiguous cases to expert radiologists [18, 19, 20]. While prior selective studies focus primarily on abstention rules, the interaction between loss-induced calibration distortion and downstream referral triage remains unexplored. We utilize selective prediction to examine clinical trade-offs between calibrated risk estimation and screening triage.

1.2. Contributions

To resolve algorithmic loss bias from distribution shift across cohort strata, this paper provides:

1. **Controlled 2×2 Experimental Design:** We evaluate two clinical architectures (DenseNet-121, ResNet-50) and two loss objectives (standard vs. positive-weighted BCE) across three independent seeds (12 models total) on 40,523 radiographs from BRAX [2], tracking discrimination and calibration across three held-out deidentified-date strata (T_1, T_2, T_3) with strict patient isolation.

- 54 **2. Mathematical Derivation and Empirical Clustering of Logit**
 55 **Shift:** We prove that positive-weighted BCE introduces a constant
 56 population logit offset $\ln w$ (Section S1 of the Supplementary Material).
 57 Across 16,604 evaluations, network logits cluster tightly around theoreti-
 58 cal $\ln w$ (mean $\Delta z = 3.29 \pm 0.81$ vs. theoretical 3.07 for Pleural Effusion).
 59 We show that temperature scaling structurally fails to correct this inter-
 60 cept displacement ($\alpha \approx -2.96$), whereas a closed-form analytic offset
 61 ($z - \ln w$) eliminates 73.7% of stratum T_3 Brier error ($0.1079 \rightarrow 0.0284$)
 62 and restores calibration parity without model retraining.
- 63 **3. Leak-Free Cross-Fitting and Patient-Clustered Bootstrap:** We
 64 implement a 5-fold patient-clustered cross-fitting protocol on T_1 and
 65 evaluate out-of-sample generalization on T_2 and T_3 backed by 2,000-
 66 replicate patient-clustered bootstrap intervals.
- 67 **4. Operational Selective Prediction and Cross-Institutional Trans-**
 68 **portability:** We formalize budget-constrained selective deferral, quanti-
 69 fying operational trade-offs between calibrated risk and screening triage,
 70 and stress-test external transportability on MIMIC-CXR-JPG (3,403
 71 images, 289 patients) under real-world domain shift.

72 2. Materials and Methods

73 2.1. Dataset and Cohort Stratification

74 We utilize the Brazilian Chest X-ray (BRAX) v1.1.0 benchmark [2]. BRAX
 75 contains 40,967 radiographs labeled with 14 thoracic findings extracted from
 76 Portuguese reports via a validated NLP pipeline. After excluding 444 radio-
 77 graphs from 89 patients whose examinations crossed partition boundaries, the
 78 analytic cohort comprised 40,523 radiographs acquired at Hospital Israelita
 79 Albert Einstein in São Paulo, Brazil.

80 In compliance with HIPAA de-identification standards, timestamps are
 81 protected as fictitious 8-digit calendar integers (YYYYMMDD) preserving intra-
 82 patient acquisition intervals. Because cross-patient calendar synchronization is
 83 unverified, partitions represent sequential deidentified-date strata rather than
 84 synchronized calendar years. Studies are partitioned chronologically around
 85 a strict model freeze boundary ($T_{\text{freeze}} = 2015-07-01$): training studies before
 86 2013-09-01 ($N = 25,123$ images, 11,265 patients, 14,315 studies); validation
 87 studies between 2013-09-01 and 2015-06-30 ($N = 7,098$ images, 3,281 patients,
 88 4,012 studies); and held-out test studies between 2015-07-01 and 2017-12-17
 89 ($N = 8,302$ images, 3,807 patients, 4,670 studies) partitioned into three

Table 1: Cohort Demographics, Radiographic Projections, and Target Condition Prevalence across Stratified Cohorts

Cohort Stratum	Images (N)	Patients (N)	Studies (N)	Sex (% M / F)	Age (Mean $\pm$ SD)	Views (% PA / AP / Lat / Unk)	Pleural Effusion N (%)	Cardiomegaly N (%)	Pneumonia N (%)	Edema N (%)
Train (Pre-deployment)	25,123	11,265	14,315	49.3% / 50.7%	46.0 $\pm$ 24.3	34.0% / 13.5% / 0.0% / 27.2%	1,116 (4.44%)	2,398 (9.55%)	456 (1.82%)	23 (0.09%)
Val (Pre-deployment)	7,098	3,281	4,012	49.8% / 50.2%	46.2 $\pm$ 24.1	34.7% / 12.2% / 0.0% / 27.7%	295 (4.16%)	681 (9.59%)	128 (1.80%)	8 (0.11%)
Test T1 (2015)	1,879	913	1,039	47.8% / 52.2%	44.5 $\pm$ 23.9	34.4% / 10.2% / 0.0% / 29.0%	99 (5.27%)	147 (7.82%)	50 (2.66%)	4 (0.21%)
Test T2 (2016)	3,987	1,832	2,249	48.5% / 51.5%	46.1 $\pm$ 23.8	34.6% / 12.0% / 0.0% / 27.6%	177 (4.44%)	422 (10.58%)	79 (1.98%)	13 (0.33%)
Test T3 (2017)	2,436	1,164	1,382	50.9% / 49.1%	46.0 $\pm$ 24.6	34.4% / 12.4% / 0.0% / 28.1%	85 (3.49%)	258 (10.59%)	50 (2.05%)	0 (0.00%)
Test All (Pooled)	8,302	3,807	4,670	49.0% / 51.0%	45.7 $\pm$ 24.1	34.5% / 11.7% / 0.0% / 28.1%	361 (4.35%)	827 (9.96%)	179 (2.16%)	17 (0.20%)

consecutive strata: T_1 ($N = 1,879$), T_2 ($N = 3,987$), and T_3 ($N = 2,436$).
 Strict patient isolation is enforced ($\text{Patients}_{\text{train}} \cap \text{Patients}_{\text{val}} \cap \text{Patients}_{\text{test}} = \emptyset$).
 Complete cohort characteristics are summarized in Table 1.

2.2. External Stress-Testing Cohort (MIMIC-CXR-JPG)

To assess whether the mathematical calibration mechanism survives cross-institutional domain shift, frozen BRAX models were externally evaluated on all eligible frontal radiographs (AP and PA views) from the official patient-partitioned MIMIC-CXR-JPG test split [21, 22]. The cohort comprises $N = 3,403$ frontal radiographs ($N = 3,041$ diagnostic studies across $N = 289$ unique patients; AP: 2,405 [70.67%], PA: 998 [29.33%]). Ground-truth labels were extracted using the official CheXpert labeler under the uncertainty-zero (U -zero) policy: Pleural Effusion 32.18% ($N = 1,095$ positive radiographs across 178 positive and 273 negative patients; 990 studies), Cardiomegaly 26.33% ($N = 896$ across 183 positive and 282 negative patients; 808 studies), Edema 21.33% ($N = 726$ across 150 positive and 287 negative patients; 659 studies), and Pneumonia 10.05% ($N = 342$ across 137 positive and 286 negative patients; 309 studies). Patient-clustered non-parametric bootstrap inference ($B = 1,000$ replicates) accounts for repeated within-patient examinations and longitudinal disease transitions across serial studies. No model fine-tuning or retraining was performed.

2.3. Architectures and Training Objectives

We evaluate two ImageNet-initialized clinical architectures: DenseNet-121 [23] (dense feature connectivity, standard in chest radiography [1]) and ResNet-50 [24] (residual skip connections as a structural comparator). Each is trained under two objectives:

1. Unweighted Binary Cross-Entropy (Standard BCE):

$$\mathcal{L}_{\text{std}}(y, \hat{p}) = -[y \log \hat{p} + (1 - y) \log(1 - \hat{p})] \quad (1)$$

where $\hat{p} = \sigma(z) = 1/(1 + e^{-z})$ is the sigmoid probability from raw logit z .

118 2. Positive-Weighted Binary Cross-Entropy (Weighted BCE):

$$\mathcal{L}_{\text{pos}}(y, \hat{p}) = -[w \cdot y \log \hat{p} + (1 - y) \log(1 - \hat{p})] \quad (2)$$

119 where $w = (N_{\text{total}} - N_{\text{pos}})/N_{\text{pos}}$ is the inverse positive training prevalence
 120 ($w = 21.51$ for Pleural Effusion, $w = 9.48$ for Cardiomegaly, $w = 54.09$
 121 for Pneumonia).

122 Models were trained using AdamW (learning rate 10^{-4} , weight decay
 123 10^{-2}) with Cosine Annealing across 15 epochs (batch size 64, AMP fp16,
 124 512×512). Pleural Effusion was pre-specified as the primary target due to
 125 stable evaluability across all held-out strata ($T_1 = 1,879, T_2 = 3,987, T_3 =$
 126 $2,436$); Cardiomegaly, Pneumonia, and Edema served as secondary validation
 127 targets. Checkpoints were selected by validation AUROC across three random
 128 seeds ($S \in \{42, 123, 2026\}$; 12 models total). Three-seed ensembles averaged
 129 probabilities: $\bar{p}_{\text{raw}}(x) = \frac{1}{3} \sum_{s \in S} \sigma(z_s(x))$ and $\bar{p}_{\text{corrected}}(x) = \frac{1}{3} \sum_{s \in S} \sigma(z_s(x) -$
 130 $\ln w)$. The empirical logit offset was evaluated across all 16,604 ensemble
 131 evaluations ($8,302 \text{ images} \times 2 \text{ architectures}$; $49,812 \text{ seed pairs}$):

$$\Delta z(x) = \bar{z}_w(x) - \bar{z}_u(x) = \frac{1}{3} \sum_{s \in S} (z_{w,s}(x) - z_{u,s}(x)) \quad (3)$$

132 2.4. Mathematical Mechanics of Loss Weighting and Calibration

133 Under positive-weighted binary cross-entropy, the conditional expected
 134 risk with positive weight $w > 0$ is:

$$\mathbb{E}_{Y|X}[\mathcal{L}_{\text{pos}}(Y, \hat{p})] = -w \cdot P(Y = 1 | X) \ln \hat{p} - P(Y = 0 | X) \ln(1 - \hat{p}) \quad (4)$$

135 Setting the first-order condition $\frac{\partial}{\partial z} \mathbb{E}[\mathcal{L}_{\text{pos}}] = 0$ yields the optimal predicted
 136 probability:

$$\hat{p}^*(x) = \frac{w \cdot P(Y = 1 | X = x)}{w \cdot P(Y = 1 | X = x) + P(Y = 0 | X = x)} \quad (5)$$

137 Converting to log-odds yields the Bayes-optimal logit:

$$z^*(x) = \ln \left(\frac{P(Y = 1 | X = x)}{P(Y = 0 | X = x)} \right) + \ln w = z_{\text{true}}(x) + \ln w \quad (6)$$

138 (see Section S1 of the Supplementary Material for the complete algebraic
 139 proof). Equation (6) derives that positive loss weighting shifts the model's
 140 population Bayes-optimal logit by $+\ln w$, connecting classical choice-based
 141 sampling corrections [13] with the class-weight likelihood recovery formulation
 142 of Caplin et al. [14]. In finite neural networks, empirical logits fluctuate
 143 around this theoretical attractor.

144 2.4.1. Limitation of Temperature Scaling

145 Temperature scaling [15] adjusts probabilities via a single scalar T : $p_{\text{temp}} =$
 146 $\sigma(z/T)$. In the Cox calibration framework [6, 7]:

$$\text{logit}(P(Y = 1 \mid X)) = \alpha + \beta \cdot \text{logit}(\hat{p}) \quad (7)$$

147 where $\alpha = 0$ represents ideal calibration-in-the-large and $\beta = 1$ denotes
 148 ideal slope. Because positive weighting induces $z \approx z_{\text{true}} + \ln w$, raw logits
 149 suffer acute intercept discrepancy ($\alpha \approx -\ln w \ll 0$). Temperature scaling
 150 adjusts slope ($\beta \rightarrow \beta \cdot T$) by multiplying logits by $1/T$, but cannot shift the
 151 additive intercept α . In low-prevalence domains dominated by negative cases,
 152 optimization selects $T^* \approx 1.0$, leaving the underlying intercept displacement
 153 uncorrected.

154 2.4.2. Post-Hoc Recalibration Suite and Computational Complexity

155 We evaluate four post-hoc recalibration functions:

- 156 1. **Temperature Scaling (T^*):** $p = \sigma(z/T^*)$, optimized via bounded
 157 Brent minimization ($O(N_{\text{val}})$ fitting, $O(1)$ inference).
- 158 2. **Analytic Offset ($z - \ln w$):** Subtracts the theoretical logit shift ($p =$
 159 $\sigma(z - \ln w)$), requiring zero validation data or tuning ($O(1)$ fitting and
 160 inference) and resetting $\alpha \approx 0$ while preserving slope.
- 161 3. **Fitted Offset ($z + a^*$):** Optimizes scalar intercept a^* on validation
 162 log-likelihood with slope fixed at 1.0 ($O(N_{\text{val}})$ fitting, $O(1)$ inference).
- 163 4. **Platt Scaling ($a + bz$):** Logistic regression on logits with non-negative
 164 slope constraint $b \geq 0$ ($O(k \cdot N_{\text{val}})$ fitting, $O(1)$ inference).

165 2.5. Cross-Fitting and Resampling Evaluation Protocol

166 On deployment stratum T_1 (2015-H2), calibrator parameters are estimated
 167 using **5-fold patient-clustered cross-fitting** (all studies from any given
 168 patient assigned to a single fold). Calibrators fit on T_1 are frozen and
 169 applied strictly out-of-sample on later strata T_2 (2016) and T_3 (2017). All
 170 95% confidence intervals are computed using $B = 2,000$ patient-clustered
 171 bootstrap resamples with replacement ($B = 1,000$ for external MIMIC
 172 validation). Paired calibrator contrasts are evaluated on identical bootstrap
 173 resamples.

174 2.6. Selective Prediction and Deferral Policy

175 In operational deployment, models can selectively defer ambiguous ex-
 176 aminations to expert radiologists [18, 19, 20]. Rather than developing novel
 177 rejection algorithms, our selective prediction formulation evaluates how loss
 178 weighting and calibration impact clinical referral efficiency and automated
 179 error rates.

180 Operating decision cutoffs t^* are determined strictly on development
 181 stratum T_1 using Youden’s J index ($t^* \approx 0.014$ for unweighted, $t^* \approx 0.022$ for
 182 analytic offset, $t^* \approx 0.33$ for raw weighted models). We define a threshold-
 183 relative uncertainty metric based on piecewise distance from t^* :

$$u(x) = 1 - \frac{|\hat{p}(x) - t^*|}{\mathbb{I}(\hat{p}(x) \geq t^*)(1 - t^*) + \mathbb{I}(\hat{p}(x) < t^*)t^*} \in [0, 1] \quad (8)$$

184 where $u(x) \rightarrow 1$ indicates maximal ambiguity ($\hat{p}(x) \approx t^*$) and $u(x) \rightarrow 0$
 185 indicates decisive confidence. At coverage budget $C \in [0.5, 1.0]$, the system
 186 retains the fraction C of cases with lowest uncertainty ($N_{\text{auto}} = \lceil C \cdot N \rceil$) and
 187 refers the remaining $1 - C$ to radiologist review ($N_{\text{ref}} = N - N_{\text{auto}}$).

188 We track triage outcomes across five distinct metrics: (i) retained pos-
 189 itives ($N_{\text{pos, ret}}$); (ii) automated true positives ($TP_{\text{auto}} = \sum_{i \in \text{Auto}} \mathbb{I}(y_i =$
 190 $1 \wedge \hat{y}_i = 1)$); (iii) automated false negatives ($FN_{\text{auto}} = \sum_{i \in \text{Auto}} \mathbb{I}(y_i =$
 191 $1 \wedge \hat{y}_i = 0)$, where $TP_{\text{auto}} + FN_{\text{auto}} = N_{\text{pos, ret}}$); (iv) retained sensitivity
 192 ($\text{Sens}_{\text{ret}} = TP_{\text{auto}}/N_{\text{pos, ret}}$); and (v) full-cohort triage sensitivity ($\text{Sens}_{\text{triage}} =$
 193 $1 - FN_{\text{auto}}/N_{\text{pos, total}}$), assuming deferred cases are correctly diagnosed by
 194 expert radiologists.

195 3. Experimental Results

196 3.1. Experimental Matrix: Discrimination vs. Calibration Trajectories

197 Table 2 presents the comprehensive evaluation of the 2×2 experimental
 198 design comprising two architectures and two loss objectives (with three
 199 independently trained random seed replications per configuration, 12 models
 200 total) for Pleural Effusion, Cardiomegaly, and Pneumonia across chronological
 201 strata T_1, T_2, T_3 .

202 3.1.1. Discrimination Attenuation vs. Calibration Trajectories

203 Both architectures exhibit moderate discriminatory attenuation across
 204 subsequent deidentified-date cohort strata. For DenseNet-121 on Pleural

Algorithm 1 Budget-Constrained Uncertainty-Ranked Selective Deferral Pipeline

Require: Cohort radiographs $\{x_i\}_{i=1}^N$, trained neural network f_θ , positive loss weight w , clinical operating threshold t^* , target coverage budget $C \in (0, 1]$.

Ensure: Automated decisions $\{\hat{y}_i : i \in \mathcal{S}\}$ and referred examination set $\mathcal{D}$.

```
1: Phase 1: Analytic Recalibration and Uncertainty Scoring
2: for  $i = 1$  to  $N$  do
3:   Compute raw network logit:  $z_i \leftarrow f_\theta(x_i)$ 
4:   if model was trained with positive class weight  $w > 1$  then
5:     Apply constant-time closed-form offset:  $\tilde{z}_i \leftarrow z_i - \ln(w)$ 
6:   else
7:      $\tilde{z}_i \leftarrow z_i$ 
8:   end if
9:   Compute calibrated probability:  $\hat{p}_i \leftarrow \sigma(\tilde{z}_i) = \frac{1}{1 + \exp(-\tilde{z}_i)}$ 
10:  Compute threshold-relative normalizer:
11:     $s_i \leftarrow \begin{cases} 1 - t^*, & \text{if } \hat{p}_i \geq t^* \\ t^*, & \text{otherwise} \end{cases}$ 
12:  Compute normalized threshold-proximity uncertainty:
13:     $u_i \leftarrow 1 - \min\left(1, \frac{|\hat{p}_i - t^*|}{s_i}\right) \in [0, 1]$ 
14: end for
15: Phase 2: Budget-Constrained Triage Partitioning
16: Compute automation quota:  $k \leftarrow \min(N, \max(1, \lceil C \cdot N \rceil))$ 
17: Sort cohort indices by ascending uncertainty:  $\pi \leftarrow \text{argsort}(\{u_1, \dots, u_N\})$ 
18: Partition cohort into automated and deferred sets:
19:    $\mathcal{S} \leftarrow \{\pi(1), \pi(2), \dots, \pi(k)\}$    {Retained automated cohort}
20:    $\mathcal{D} \leftarrow \{\pi(k+1), \dots, \pi(N)\}$    {Referred to expert radiologist review}
21: Phase 3: Decision Assignment
22: for  $i \in \mathcal{S}$  do
23:   Assign automated binary classification:  $\hat{y}_i \leftarrow \mathbb{I}(\hat{p}_i \geq t^*)$ 
24: end for
25: return Automated decisions  $\{\hat{y}_i\}_{i \in \mathcal{S}}$ , deferred cases  $\mathcal{D}$ .
```

205 Effusion, AUROC decreases from 0.9335 ± 0.0052 at T_1 to 0.8731 ± 0.0120 at T_3
206 ($\Delta\text{AUROC} = -0.0604$); ResNet-50 exhibits an identical downward trajectory
207 ($0.9295 \pm 0.0113 \rightarrow 0.8608 \pm 0.0190$, $\Delta\text{AUROC} = -0.0687$; Table 2). This

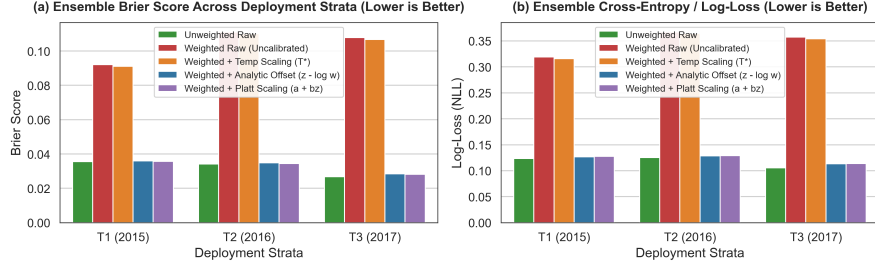


Figure 1: Multi-calibrator evaluation on Pleural Effusion for the DenseNet-121 3-seed ensemble across sequential deidentified-date strata (T_1, T_2, T_3), directly visualizing Table 3. (a) Brier score and (b) Log-loss. Temperature scaling fails to reduce probability error in weighted models, whereas Analytic Offset ($z - \log w$) and Platt Scaling reduce Brier score by 73.7% ($\Delta\text{Brier} = -0.0795$), matching unweighted models.

confirms that discriminatory attenuation occurs gradually across subsequent cohort strata regardless of loss objective.

In stark contrast, probability calibration diverges sharply by loss objective. For unweighted models, Brier scores are low and actually improve as prevalence falls ($0.0369 \pm 0.0029 \rightarrow 0.0279 \pm 0.0009$ on Pleural Effusion). For positive-weighted models, Brier scores are severely elevated and appear to deteriorate over time ($0.1081 \pm 0.0394 \rightarrow 0.1254 \pm 0.0456$ on DenseNet-121; $0.1256 \pm 0.0325 \rightarrow 0.1392 \pm 0.0310$ on ResNet-50), with ECE exceeding 22%. Tracking the excess penalty $\Delta\text{Brier} = \text{Brier}_{\text{weighted}} - \text{Brier}_{\text{unweighted}}$ across sequential strata reveals the operational mechanism: when prevalence declines (Pleural Effusion: 5.27% $\rightarrow$ 3.49%), the excess penalty expands ($+0.0712 \rightarrow +0.0975$); when prevalence rises (Cardiomegaly: 7.82% $\rightarrow$ 10.59%), it attenuates ($+0.0729 \rightarrow +0.0683$); and when prevalence is steady (Pneumonia: 2.66% $\rightarrow$ 2.05%), it remains fixed ($+0.1079 \rightarrow +0.1112$). A static training logit shift produces surplus false-positive probability mass that is penalized more severely when deployment disease prevalence drops, expanding the apparent temporal calibration gap.

3.2. Multi-Calibrator Comparison

Table 3 and Fig. 1 benchmark post-hoc calibrators on Pleural Effusion across 2,000 patient-clustered bootstrap resamples on the 3-seed ensemble.

For weighted DenseNet-121, temperature scaling optimizes $T^* \approx 1.0$, producing an uncalibrated T_3 Brier score of 0.1067 compared to raw 0.1079 ($\Delta\text{Brier} = -0.0011$ [95% CI: $-0.0014, -0.0009$]) while leaving the calibration

Table 2: Experimental Matrix Evaluation: Primary Longitudinal Performance across Architectures, Loss Objectives, and Thoracic Pathologies (Strata $T_1 \rightarrow T_3$). Values represent mean $\pm$ standard deviation across three independently trained random seeds ($S \in \{42, 123, 2026\}$). Complete 17-column trajectories across all intermediate deployment strata (T_1, T_2, T_3) and log-loss metrics are detailed in Supplementary Table S3.

Architecture	Objective	Pathology	AUROC ($T_1 \rightarrow T_3$)	Δ AUROC	Brier ($T_1 \rightarrow T_3$)	Δ Brier	ECE T_3
DenseNet-121	Unweighted BCE	Pleural Effusion	0.934 $\rightarrow$ 0.873	-0.060	0.037 $\rightarrow$ 0.028	-0.009	0.0090
DenseNet-121	Unweighted BCE	Cardiomegaly	0.908 $\rightarrow$ 0.902	-0.006	0.053 $\rightarrow$ 0.068	+0.015	0.0210
DenseNet-121	Unweighted BCE	Pneumonia	0.847 $\rightarrow$ 0.865	+0.017	0.024 $\rightarrow$ 0.019	-0.005	0.0095
DenseNet-121	Unweighted BCE	Edema*	0.540 $\rightarrow$ -	-	0.002 $\rightarrow$ 0.000	-0.002	0.0029
DenseNet-121	Pos-Weighted BCE	Pleural Effusion	0.906 $\rightarrow$ 0.839	-0.067	0.108 $\rightarrow$ 0.125	+0.017	0.2227
DenseNet-121	Pos-Weighted BCE	Cardiomegaly	0.874 $\rightarrow$ 0.858	-0.016	0.126 $\rightarrow$ 0.136	+0.010	0.1969
DenseNet-121	Pos-Weighted BCE	Pneumonia	0.818 $\rightarrow$ 0.842	+0.024	0.131 $\rightarrow$ 0.130	-0.002	0.2425
DenseNet-121	Pos-Weighted BCE	Edema*	0.612 $\rightarrow$ -	-	0.061 $\rightarrow$ 0.064	+0.002	0.1274
ResNet-50	Unweighted BCE	Pleural Effusion	0.930 $\rightarrow$ 0.861	-0.069	0.037 $\rightarrow$ 0.028	-0.009	0.0132
ResNet-50	Unweighted BCE	Cardiomegaly	0.890 $\rightarrow$ 0.879	-0.011	0.057 $\rightarrow$ 0.071	+0.014	0.0175
ResNet-50	Unweighted BCE	Pneumonia	0.843 $\rightarrow$ 0.880	+0.037	0.024 $\rightarrow$ 0.019	-0.005	0.0106
ResNet-50	Unweighted BCE	Edema*	0.556 $\rightarrow$ -	-	0.002 $\rightarrow$ 0.000	-0.002	0.0017
ResNet-50	Pos-Weighted BCE	Pleural Effusion	0.879 $\rightarrow$ 0.819	-0.060	0.126 $\rightarrow$ 0.139	+0.014	0.2751
ResNet-50	Pos-Weighted BCE	Cardiomegaly	0.830 $\rightarrow$ 0.799	-0.031	0.142 $\rightarrow$ 0.152	+0.011	0.2338
ResNet-50	Pos-Weighted BCE	Pneumonia	0.835 $\rightarrow$ 0.823	-0.012	0.167 $\rightarrow$ 0.169	+0.002	0.3253
ResNet-50	Pos-Weighted BCE	Edema*	0.549 $\rightarrow$ -	-	0.069 $\rightarrow$ 0.075	+0.005	0.1670

Table 3: Multi-Calibrator Comparison on Pleural Effusion: Deidentified-Date Ensemble Performance, Calibration Intercept (α), Slope (β), and Paired Bootstrap Contrasts. Complete 12-column factorial evaluation across all intermediate strata and log-loss is provided in Supplementary Table S4.

Architecture	Loss Objective	Calibrator	Brier ($T_1 \rightarrow T_3$)	T_3 α	T_3 β	T_3 ECE	T_3 Δ Brier vs Raw [95% CI]
DenseNet-121	Weighted BCE	Raw (Uncalibrated)	0.092 $\rightarrow$ 0.108	-2.964	1.048	0.2227	-
DenseNet-121	Weighted BCE	Temperature Scaling (T^*)	0.091 $\rightarrow$ 0.107	-2.962	1.029	0.2190	-0.0011 [-0.0014, -0.0009]
DenseNet-121	Weighted BCE	Analytic Offset ($z - \log w$)	0.036 $\rightarrow$ 0.028	-0.445	0.929	0.0126	-0.0795 [-0.0893, -0.0697]
DenseNet-121	Weighted BCE	Platt Scaling ($a + bz$)	0.036 $\rightarrow$ 0.028	-0.229	1.128	0.0176	-0.0798 [-0.0894, -0.0702]
DenseNet-121	Unweighted BCE	Raw Baseline	0.036 $\rightarrow$ 0.027	+0.147	1.044	0.0040	-
ResNet-50	Weighted BCE	Raw (Uncalibrated)	0.110 $\rightarrow$ 0.123	-2.853	1.950	0.2751	-
ResNet-50	Weighted BCE	Temperature Scaling (T^*)	0.088 $\rightarrow$ 0.101	-2.608	1.184	0.1993	-0.0220 [-0.0241, -0.0198]
ResNet-50	Weighted BCE	Analytic Offset ($z - \log w$)	0.040 $\rightarrow$ 0.029	+1.075	1.405	0.0158	-0.0936 [-0.1022, -0.0844]
ResNet-50	Weighted BCE	Platt Scaling ($a + bz$)	0.038 $\rightarrow$ 0.030	-0.191	1.178	0.0223	-0.0930 [-0.1007, -0.0849]
ResNet-50	Unweighted BCE	Raw Baseline	0.036 $\rightarrow$ 0.027	-0.077	1.095	0.0109	-

intercept severely displaced ($\alpha = -2.9615$ vs. raw -2.9638). For ResNet-50, temperature scaling achieves modest variance shrinkage (Δ Brier = -0.0220 [$-0.0241, -0.0198$]), but leaves the underlying intercept discrepancy uncorrected ($\alpha = -2.6080$ vs. raw -2.8526). Temperature scaling is mathematically incapable of adjusting this intercept displacement.

In contrast, subtracting the theoretical logit offset ($z - \ln w$, with $\ln(21.51) = +3.068$) drops T_3 Brier from 0.1079 to 0.0284 (Δ Brier = -0.0795 [95% CI: $-0.0893, -0.0697$], 73.7% error reduction), reducing ECE from 22.3% to 1.26% and shifting the Cox intercept from $\alpha = -2.9638$ to -0.4451 while perfectly preserving the calibration slope ($\beta = 0.9292$). Fitted Offset ($z + a^*$)

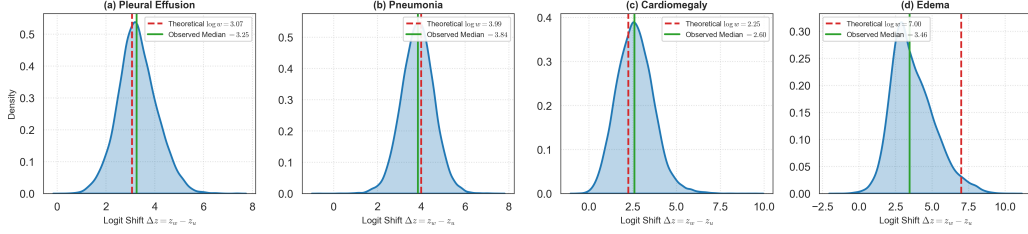


Figure 2: Empirical verification of theoretical logit shift ($z_w - z_u \approx \log w$) across 16,604 held-out deidentified-date image evaluations (8,302 images $\times$ 2 architectures, with Δz computed as the mean difference across the three seed-matched weighted/unweighted model pairs). Kernel density estimates (blue) of observed logit differences between positive-weighted and unweighted models cluster tightly around the theoretical population attractor $\log w$ (dashed red line; observed median in solid green) across Pleural Effusion (a), Pneumonia (b), and Cardiomegaly (c). For Edema (d), extreme training class scarcity ($w > 1000$) induces optimization saturation, bounding empirical shifts near $+3.69$.

241 and Platt Scaling achieve identical performance (T_3 Brier: 0.0289 and 0.0281;
 242 $\Delta\text{Brier} = -0.0790$ and -0.0798). For ResNet-50, Analytic Offset similarly
 243 eliminates 76.3% of probability error ($0.1227 \rightarrow 0.0291$, $\Delta\text{Brier} = -0.0936$
 244 [95% CI: $-0.1022, -0.0844$]), confirming that intercept neutralization is ar-
 245 chitecturally universal.

246 3.2.1. Empirical Verification of Logit Offset: $\Delta z \approx \ln w$

247 Evaluating logit differences $\Delta z(x) = \bar{z}_w(x) - \bar{z}_u(x)$ across all 16,604
 248 ensemble evaluations (Fig. 2) confirms that finite neural networks cluster
 249 tightly around theoretical $\ln w$: Pleural Effusion ($\ln w = 3.07$) observed mean
 250 $\Delta z = 3.29 \pm 0.81$ (median 3.25); Pneumonia ($\ln w = 3.99$) observed 3.83 ± 0.74
 251 (median 3.84); and Cardiomegaly ($\ln w = 2.25$) observed 2.64 ± 1.04 (median
 252 2.60). For Edema ($w = 1091.3, \ln w = 7.00$), acute training scarcity (23
 253 positive cases) causes optimization saturation, bounding empirical shifts
 254 at $\Delta z = 3.69 \pm 1.47$ (median 3.46) due to L_2 regularization. Equal-mass
 255 reliability diagrams (Fig. 3) demonstrate that the analytic offset realigns
 256 curves tightly with the ideal diagonal across all strata.

257 3.3. Selective Prediction and Workload Triage Simulation

258 Table 4 and Fig. 4 summarize the selective triage simulation on later
 259 held-out stratum T_3 ($N = 2,436$ radiographs, 85 positive Pleural Effusion
 260 cases) under development thresholds. At 100% coverage (full automation),
 261 the raw weighted model incurs 16.0 missed cases (FN_{auto}) with Brier 0.1079.

Table 4: Selective Prediction and Reader Referral Triage Simulation on Later Held-Out Stratum T_3 ($N = 2,436$ Radiographs, 85 Positive Pleural Effusion Cases). Evaluated under matched Youden decision thresholds determined on development stratum T_1 across primary clinical operating regimes (100%, 90%, 80%, 70% coverage). Complete 24-row simulation across all coverage levels is detailed in Supplementary Table S5.

Coverage	Referred (Workload)	Strategy	Automated FN (Missed)	Retained Sens.	Full Sens.	Retained AUROC	Retained Brier
100%	0 (0%)	Weighted BCE (Raw)	16.0	81.18%	81.18%	0.8691	0.1079
100%	0 (0%)	Weighted BCE (Analytic Corrected)	16.0	81.18%	81.18%	0.8623	0.0284
100%	0 (0%)	Unweighted BCE (Raw)	12.0	85.88%	85.88%	0.8806	0.0268
90%	243 (10%)	Weighted BCE (Raw)	14.0	82.93%	83.53%	0.8789	0.1062
90%	243 (10%)	Weighted BCE (Analytic Corrected)	15.0	81.01%	82.35%	0.8735	0.0289
90%	243 (10%)	Unweighted BCE (Raw)	12.0	84.62%	85.88%	0.8923	0.0266
80%	487 (20%)	Weighted BCE (Raw)	10.0	87.01%	88.24%	0.8882	0.1012
80%	487 (20%)	Weighted BCE (Analytic Corrected)	15.0	77.61%	82.35%	0.8896	0.0264
80%	487 (20%)	Unweighted BCE (Raw)	12.0	82.61%	85.88%	0.9027	0.0256
70%	730 (30%)	Weighted BCE (Raw)	7.0	89.39%	91.76%	0.8997	0.0956
70%	730 (30%)	Weighted BCE (Analytic Corrected)	13.0	70.45%	84.71%	0.8876	0.0178
70%	730 (30%)	Unweighted BCE (Raw)	11.0	78.00%	87.06%	0.9051	0.0194
70%	730 (30%)	Random Deferral Baseline	11.3	80.99%	86.69%	—	0.1081

262 Calibrated Analytic Offset under matched threshold ($t^* \approx 0.022$) yields
 263 identical sensitivity (81.18%) and 16.0 missed cases, but with an almost
 264 four-fold lower Brier score of 0.0284. Unweighted raw models achieve 12.0
 265 missed cases (sensitivity 85.88%, Brier 0.0268).

266 Under selective deferral at 70% coverage (30% referral workload, 730
 267 radiographs referred): raw weighted models halve automated false negatives
 268 from 16.0 to 7.0 (retained sensitivity 89.39%, full-cohort triage sensitivity
 269 91.76%), but leave probabilities uncalibrated (Brier 0.0956); conversely, ana-
 270 lytic calibrated models achieve superior probability estimation (Brier 0.0178),
 271 but retain 13.0 automated false negatives because threshold-distance nor-
 272 malization ($t^* \approx 0.022$) classifies low-probability false negatives ($\hat{p} \ll t^*$)
 273 as confident negatives. This reveals a fundamental operational trade-off:
 274 probability calibration is required for valid risk scoring, whereas raw weighted

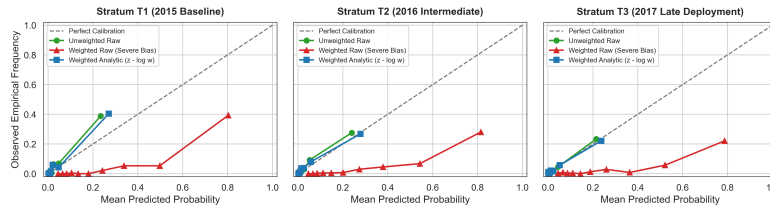


Figure 3: Reliability diagrams for Pleural Effusion (DenseNet-121 3-seed ensemble) across held-out deidentified-date strata (T_1, T_2, T_3). Raw weighted models (red) severely over-predict event probabilities across all bins. Analytic offset calibration (blue) and unweighted models (green) remain tightly calibrated to the ideal diagonal across all deployment windows.

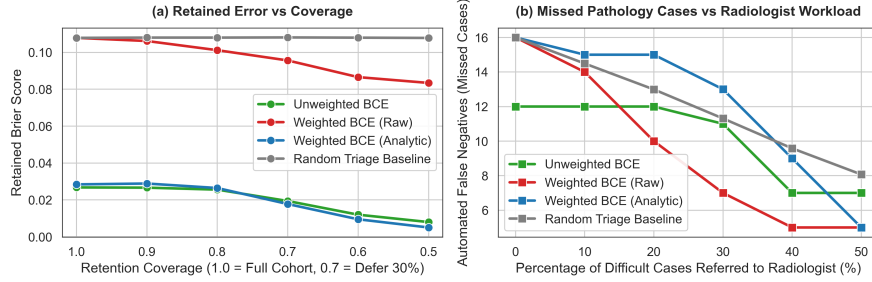


Figure 4: Selective prediction evaluation on later held-out stratum T_3 with matched development thresholds (Table 4). Left: Risk-coverage curves demonstrating that selective deferral steadily reduces Brier error across all models as ambiguous cases are referred. Right: Reader referral workload vs. automated false negatives, illustrating the distinct operational behaviors of raw weighted models (halving missed cases to 7.0 under 30% referral) and analytic corrected models (optimizing probability accuracy with Brier 0.0178).

models with threshold-proximity triage act as a heuristic for recall-oriented screening. Neither system eliminates missed disease entirely.

3.4. Cross-Institutional External Evaluation (MIMIC-CXR-JPG)

The external MIMIC-CXR-JPG cohort comprised 3,403 frontal radiographs from 3,041 studies and 289 patients. Rather than asserting turnkey diagnostic transportability—given moderate discrimination for Pleural Effusion (AUROC 0.688 [95% CI: 0.656, 0.717] DenseNet-121; 0.677 ResNet-50), weak discrimination for Pneumonia (0.583–0.607) and Edema (0.516–0.581), and near-chance discrimination for Cardiomegaly (0.519–0.525)—this experiment functions as an external stress-test of zero-retraining recalibration under domain shift. Negative Brier skill scores (BSS < 0 ; Table 5) confirmed substantial cross-institutional baseline calibration shift relative to a local null model ($\text{Brier}_{\text{null}} = \pi(1 - \pi)$).

Paired analytic recalibration ($\bar{p}_{\text{corr}} = \frac{1}{3} \sum \sigma(z_s - \ln w)$) demonstrated statistically significant Brier reductions for Cardiomegaly ($\Delta\text{Brier} = -0.0579$ [95% CI: $-0.0790, -0.0349$] DenseNet; -0.0611 [$-0.0818, -0.0415$] ResNet) and Pneumonia ($\Delta\text{Brier} = -0.0095$ [$-0.0170, -0.0016$] DenseNet; -0.0155 [$-0.0254, -0.0059$] ResNet), with point-estimate reductions on Pleural Effusion (-0.0084 and -0.0109). For Edema, subtracting theoretical $\ln w$ worsened Brier score ($+0.0217$), consistent with over-correction when empirical finite-network shifts saturate below theoretical $\ln w$. Diagnostic study-level sensitivity analysis ($N = 3,041$ studies across 289 patients, maximum proba-

Table 5: Cross-Institutional External Validation on the Official MIMIC-CXR-JPG Frontal Test Cohort ($N = 3,403$ Images from 3,041 Studies Across $N = 289$ Unique Patients). Primary experimental evaluation of paired zero-parameter analytic logit recalibration ($\bar{p}_{\text{raw}} = \frac{1}{3} \sum \sigma(z_s) \rightarrow \bar{p}_{\text{corr}} = \frac{1}{3} \sum \sigma(z_s - \log w)$) on 3-seed ensembles trained with positive-weighted BCE. Evaluated with empirical prevalence (π), local null reference baseline (Brier_{null} = $\pi(1 - \pi)$), patient-clustered 1,000-replicate bootstrap 95% confidence intervals, Brier skill score (BSS = $1 - \text{Brier}/\text{Brier}_{\text{null}}$), and paired contrast ($\Delta\text{Brier} = \text{Brier}_{\text{corr}} - \text{Brier}_{\text{raw}}$). The complete 16-row factorial comparison with unweighted BCE baselines is detailed in Supplementary Table S1, and study-level sensitivity analysis is detailed in Supplementary Table S2.

Architecture	Pathology	Prevalence (π)	Brier _{null}	AUROC [95% CI]	Brier (Raw $\rightarrow$ Corr)	ΔBrier [95% CI]	BSS (Raw $\rightarrow$ Corr)
DenseNet-121	Pleural Effusion	32.18%	0.2182	0.668 [0.636, 0.700]	0.2536 $\rightarrow$ 0.2451	-0.0084 [-0.0401, +0.0240]	-0.162 $\rightarrow$ -0.123
DenseNet-121	Cardiomegaly	26.33%	0.1940	0.525 [0.492, 0.558]	0.2907 $\rightarrow$ 0.2328	-0.0579 [-0.0790, -0.0349]	-0.499 $\rightarrow$ -0.200
DenseNet-121	Pneumonia	10.05%	0.0904	0.598 [0.553, 0.639]	0.1078 $\rightarrow$ 0.0983	-0.0095 [-0.0170, -0.0016]	-0.192 $\rightarrow$ -0.087
DenseNet-121	Edema	21.33%	0.1678	0.581 [0.549, 0.611]	0.1914 $\rightarrow$ 0.2131	+0.0217 [+0.0118, +0.0307]	-0.141 $\rightarrow$ -0.270
ResNet-50	Pleural Effusion	32.18%	0.2182	0.620 [0.588, 0.653]	0.2665 $\rightarrow$ 0.2557	-0.0109 [-0.0417, +0.0253]	-0.221 $\rightarrow$ -0.172
ResNet-50	Cardiomegaly	26.33%	0.1940	0.522 [0.491, 0.552]	0.2822 $\rightarrow$ 0.2211	-0.0611 [-0.0818, -0.0415]	-0.455 $\rightarrow$ -0.140
ResNet-50	Pneumonia	10.05%	0.0904	0.589 [0.545, 0.629]	0.1137 $\rightarrow$ 0.0982	-0.0155 [-0.0254, -0.0059]	-0.257 $\rightarrow$ -0.086
ResNet-50	Edema	21.33%	0.1678	0.570 [0.538, 0.602]	0.1929 $\rightarrow$ 0.2130	+0.0201 [+0.0092, +0.0313]	-0.149 $\rightarrow$ -0.269

297 bility pooling; Supplementary Table S2) demonstrated identical concordance,
 298 confirming that recalibration patterns hold across radiographs and studies.

299 4. Discussion

300 4.1. Decoupling Loss Distortion from Deidentified-Date Cohort Shift

301 Our findings clarify the mechanism underlying reported “temporal calibration
 302 decay” in medical AI [8, 9]. While moderate discriminatory attenuation
 303 occurs across cohort strata (AUROC drops by ≈ 0.06 across unweighted
 304 and weighted models), perceived calibration collapse is largely an artifact
 305 of positive loss weighting interacting with fluctuating prevalence. Because
 306 standard evaluations rely on temperature scaling—which rescales dispersion
 307 without shifting intercepts—evaluators have conflated static objective bias
 308 with temporal feature drift. Neutralizing the mathematical offset ($z - \ln w$) re-
 309 solves this apparent collapse, restoring Brier parity with unweighted baselines
 310 across all held-out strata.

311 4.2. Relationship to Prior Theoretical and Clinical Literature

312 Our findings extend the theoretical framework of Caplin et al. [14] from
 313 static tasks into longitudinal clinical deployment: (i) resolving the temporal
 314 calibration decay paradox across sequential deidentified-date cohort strata; (ii)
 315 demonstrating the structural failure of temperature scaling ($\alpha \approx -2.96$); (iii)
 316 bounding finite-network optimization shifts under acute scarcity ($w > 1000$ in

317 Edema saturates at $\Delta z \approx 3.69$ vs. theoretical 7.00); and (iv) evaluating cross-
 318 institutional transfer from BRAX to MIMIC-CXR. Furthermore, extending
 319 multi-dataset selective CXR investigations [20], we clarify clinical trade-offs:
 320 analytic offset ($z - \ln w$) is essential for risk stratification and prognosis
 321 (restoring Brier parity [17]), whereas raw weighted models facilitate recall-
 322 oriented screening under triage constraints.

323 4.3. Architectural Universality and Operational Utility

324 Both architectures exhibit consistent behavior: identical discrimination
 325 attenuation ($\Delta \text{AUROC} = -0.0604$ vs. -0.0687), severe raw calibration
 326 intercepts ($\alpha \approx -2.96$ and -2.85), and comparable Brier reductions under
 327 Analytic Offset (73.7% and 76.3%, $\Delta \text{Brier} = -0.0795$ and -0.0936). Oper-
 328 ating in constant time ($O(1)$) with zero validation fitting and zero refitting,
 329 Analytic Offset ($z - \ln w$) provides an immediate algorithmic solution for
 330 clinical PACS pipelines.

331 4.4. Scope, Boundaries, and Methodological Limitations

332 Several methodological limitations define the boundary of these findings:

- 333 1. **Date Anonymization and Temporal Semantics:** In BRAX, dates
 334 are fictitious hashes preserving intra-patient acquisition intervals; cross-
 335 patient calendar synchronization is unverified. Partitions (T_1, T_2, T_3)
 336 reflect sequential deidentified-date strata rather than synchronized cal-
 337 endar years.
- 338 2. **Cross-Institutional Domain Gap and Modest Transportability:**
 339 External validation on MIMIC-CXR-JPG yielded modest discrimina-
 340 tion (Effusion AUROC ≈ 0.69) and near-chance discrimination for
 341 Cardiomegaly (0.52). While analytic offset reduced Brier error for
 342 Cardiomegaly and Pneumonia, it worsened Edema (+0.0217), demon-
 343 strating that zero-retraining offset correction cannot compensate for
 344 severe cross-institutional covariate shift or negative transfer.
- 345 3. **Label Extraction Noise and Retrospective Simulation:** La-
 346 bels were derived from Portuguese reports via NLP rather than multi-
 347 radiologist panel review, and evaluations were retrospective. Real-world
 348 validation requires prospective silent shadow deployment in operational
 349 hospital PACS.
- 350 4. **Architectural Scope and Scarcity Bounds:** We evaluated standard
 351 CNN backbones (DenseNet-121, ResNet-50); Vision Transformers and

vision-language models warrant future investigation. Under extreme class scarcity (Edema, $w = 1091.3$), empirical logit shifts saturate at +3.69 due to L_2 regularization, defining the empirical boundary of the asymptotic derivation.

5. Conclusion

Across 40,523 BRAX radiographs, severe probability miscalibration across held-out strata was driven by positive loss weighting rather than progressive temporal drift. Because temperature scaling cannot adjust logit intercepts, it leaves distortion intact. A constant-time closed-form analytic offset ($z - \ln w$) eliminates 73.7% of stratum T_3 Brier error ($0.1079 \rightarrow 0.0284$, $\Delta\text{Brier} = -0.0795$ [95% CI: $-0.0893, -0.0697$]) without model refitting or validation data, restoring calibration parity with unweighted baselines. In selective triage, raw weighted models halve missed cases under a 30% referral budget, whereas analytic calibration restores reliable probabilities for decision-theoretic risk stratification. Neither approach eliminates missed disease entirely, underscoring the necessity of transparent, workload-aware deployment policies in clinical deep learning.

CRedit Authorship Contribution Statement

Vishesh Panghal: Conceptualization, Methodology, Software, Validation, Formal Analysis, Investigation, Data Curation, Writing – Original Draft, Writing – Review & Editing, Visualization, Project Administration.

Declaration of Competing Interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and Code Availability

The BRAX dataset (v1.1.0) [2] is available on PhysioNet at <https://physionet.org/content/brax/1.1.0/> under the PhysioNet Credentialed Data Use Agreement. The MIMIC-CXR-JPG database (v2.1.0) [21, 22] is available on PhysioNet at <https://physionet.org/content/mimic-cxr-jpg/>

382 2.1.0/ under the PhysioNet Credentialed Data Use Agreement. Access re-
383 quires credentialed PhysioNet researcher status, completion of human subjects
384 research training, and signed data use agreements. Complete end-to-end repli-
385 cation scripts, model evaluation routines, cross-fitting recalibration pipelines,
386 selective prediction triage simulations, and tabular evaluation artifacts are
387 openly available on GitHub at [https://github.com/Vishesh-panghal/
388 temporal-brax-calibration](https://github.com/Vishesh-panghal/temporal-brax-calibration), and the repository will be permanently as-
389 signed a persistent DOI via Zenodo upon acceptance of the manuscript. No
390 credentialed patient-level images, metadata, labels, identifiers, or prediction
391 rows are redistributed; the repository contains aggregate evaluation artifacts
392 and a synthetic manifest schema example.

393 **Ethical Approval and Consent to Participate**

394 This study represents a secondary analysis of credentialed-access, de-
395 identified diagnostic imaging databases accessible under formal Data Use
396 Agreements. The primary BRAX dataset collection was approved by the
397 Institutional Review Board (IRB) of Hospital Israelita Albert Einstein (São
398 Paulo, Brazil; CAAE: 36720520.1.0000.0071) with a formal waiver of informed
399 consent. Creation and distribution of the MIMIC-CXR-JPG database was
400 approved by the Institutional Review Boards of Beth Israel Deaconess Medical
401 Center (Boston, MA) and the Massachusetts Institute of Technology (Protocol
402 #0403000206) with a waiver of informed consent. The author completed
403 mandatory human subjects research training (CITI Program certification),
404 executed approved PhysioNet Data Use Agreements for both databases, and
405 strictly complied with data governance protocols prohibiting re-identification
406 or redistribution of protected health information.

407 **Declaration of Generative AI and AI-Assisted Technologies in the** 408 **Writing Process**

409 During the preparation of this work, the author utilized generative artifi-
410 cial intelligence platforms (Google Gemini, Anthropic Claude, and OpenAI
411 Codex) exclusively for LaTeX formatting and syntax verification, analysis code
412 refactoring and modularization, grammatical refinement, and bibliographic
413 cross-referencing. Following the use of these tools, the author thoroughly
414 reviewed, verified, and edited all text, numerical outputs, and code, and takes
415 full intellectual and scientific responsibility for the integrity, validity, and
416 conclusions of the published manuscript.

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